

The Frame Becomes Real

Sunday, July 12, 2026 · 3:00 – 4:45 PM · Mr. Holcombe's house

4-motor	4	1	1 hr 45 min
frame config, assembled minus motors	kids taught CAD by a teammate	wiring bug to solve (LED polarity)	meeting duration

Recap

Class 1.5 was the day the frame stopped being sketches on paper and became something you can pick up with two hands. Nine of us were there — Trenton, Rhylan, Rebecca, Liam, Maddox, Evan, Tyler, Isaac, and Trooper — and the meeting ran 3:00 to 4:45. The night split three ways: cutting and building the PVC frame, finishing the control board, and CAD. On the frame, we did the talking before the cutting — cutter safety, what PVC we're using, and the cut list — and then cut and assembled the full TriggerFish frame in the four-motor configuration. It's fully built now, motors being the only thing left to hang on it. On the control board, we finished soldering the backboard; we still have to screw in the tether wires, and we're short a few wires. We also hit a real problem: the LED polarity is coming back reversed and we haven't figured out why yet. Get a board like this wired wrong enough and it turns into a toxic smoke machine — so instead of powering it up, we checked it with a multimeter, and we may contact the manufacturer. On CAD, Trenton ran lessons for Rhylan, Tyler, Evan, and Maddox; the group built a LEGO-block-style model in Onshape, saved into a shared "Learning CAD" folder so everybody can learn from it.

Safety

- PVC cutting: have somebody hold both ends of the pipe so the offcut can't get flung across the bench when the cut finishes.
- Measure, mark with an Expo marker, then cut. The mark comes before the blade — never cut to an eyeballed line.
- Use the first good piece as the template for every identical piece after it. Copying a known-good cut beats re-measuring.
- Know the tool's limit: the PVC cutter doesn't leave a flush, square face, so every cut needs a cleanup pass before the pipe seats.
- On the control board — wire it right or don't power it. A miswired board isn't just dead, it's a "toxic smoke machine."

Decisions made

- Build the frame in the four-motor TriggerFish configuration. Assemble the full PVC skeleton first; motors go on last.
- Cut discipline is fixed: talk through the cut list and safety first, mark with an Expo marker, cut, then template every identical piece off the first good one.
- Don't power the control board until the wiring is verified. Reversed LED polarity means a multimeter trace, and we contact the manufacturer before any smoke test.
- CAD gets taught peer-to-peer. Trenton runs the lessons, and every practice model lives in a shared "Learning CAD" folder.

Action items

Item	Owner	Due
Trace why the LED polarity reads reversed — multimeter check, contact manufacturer if the board is otherwise correct	Control-board group	by 1.6
Screw in the tether wires and source the missing wires to finish the backboard	Control-board group	by 1.6
Mount the four motors onto the assembled PVC frame	Frame group	1.6
Weigh the completed frame and run a first buoyancy test	Sunday cohort	1.6
Keep the "Learning CAD" folder going — every kid uploads their practice Onshape model	CAD group (Trenton lead)	ongoing
Start the plan to move from analog to digital control and pick a code editor	Advisor + team	by Aug 15

What we learned

- A PVC cutter is fast but not precise. It won't leave a flush, square face, so every cut needs a cleanup pass. Knowing a tool's limits is part of using it well.
- The template trick beats re-measuring. Cut one piece right, then use it as the master — re-measuring each cut is just more chances to introduce an error.
- A control board is one of the few parts that can fail dangerously. "Working" isn't the bar — verified is. Reversed polarity is a signal to stop and meter it, not to power up and hope.
- Teaching is how you find out whether you actually understand something. Trenton taught four teammates to build a model from scratch — harder than drawing it himself.

Looking ahead

Next session is about turning the skeleton into a robot that can float. The four motors get mounted onto the PVC frame, then we weigh the whole thing and run a first buoyancy test to see how close we are to neutral. The control-board group keeps chasing the LED-polarity bug, screws in the tether wires, and finishes the backboard. Looking further out, the team wants to move from analog to digital control and settle on a code editor by August 15 — the first real step toward writing the software that flies this thing.